

# Evaluation of stochastic trajectory-based epidemic models using the energy score

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## The Energy Score for Trajectory-Based Projections

- The **energy score (ES)** is a multivariate, strictly proper scoring rule applied to stochastic trajectories that evaluates a prediction based on calibration and sharpness.
- It is calculated for a distribution  $P$  and vector of observed data  $y$ , and is defined as:

$$ES(P, y) = \underbrace{\frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{m=1}^M (x_m^{(i)} - y_m)^2}}_{\text{distance from surveillance data}} - \underbrace{\frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N \sqrt{\sum_{m=1}^M (x_m^{(i)} - x_m^{(j)})^2}}_{\text{distance between trajectories}}$$

where  $N$  is the number of trajectories,  $M$  is the length of the time series, and  $x_m^{(i)}$  is the predicted value by trajectory  $i$  at time  $m$ .

- The ES is an absolute measure. It places greater weight on signals with large magnitude.
- We define a **relative ES** (right) that normalizes the score by the outbreak's total size.
- The **multi-dimensional ES** evaluates model performance across multiple projections dimensions (i.e. locations) with a single metric. It compares a  $T \times M$  matrix of predictions  $A$  with  $T$  outcomes and  $M$  time points with a surveillance data matrix  $Y$  and with normalization term  $\Phi$ . Here,  $\Phi_j = 1/\sum_m y_{jm}$ .

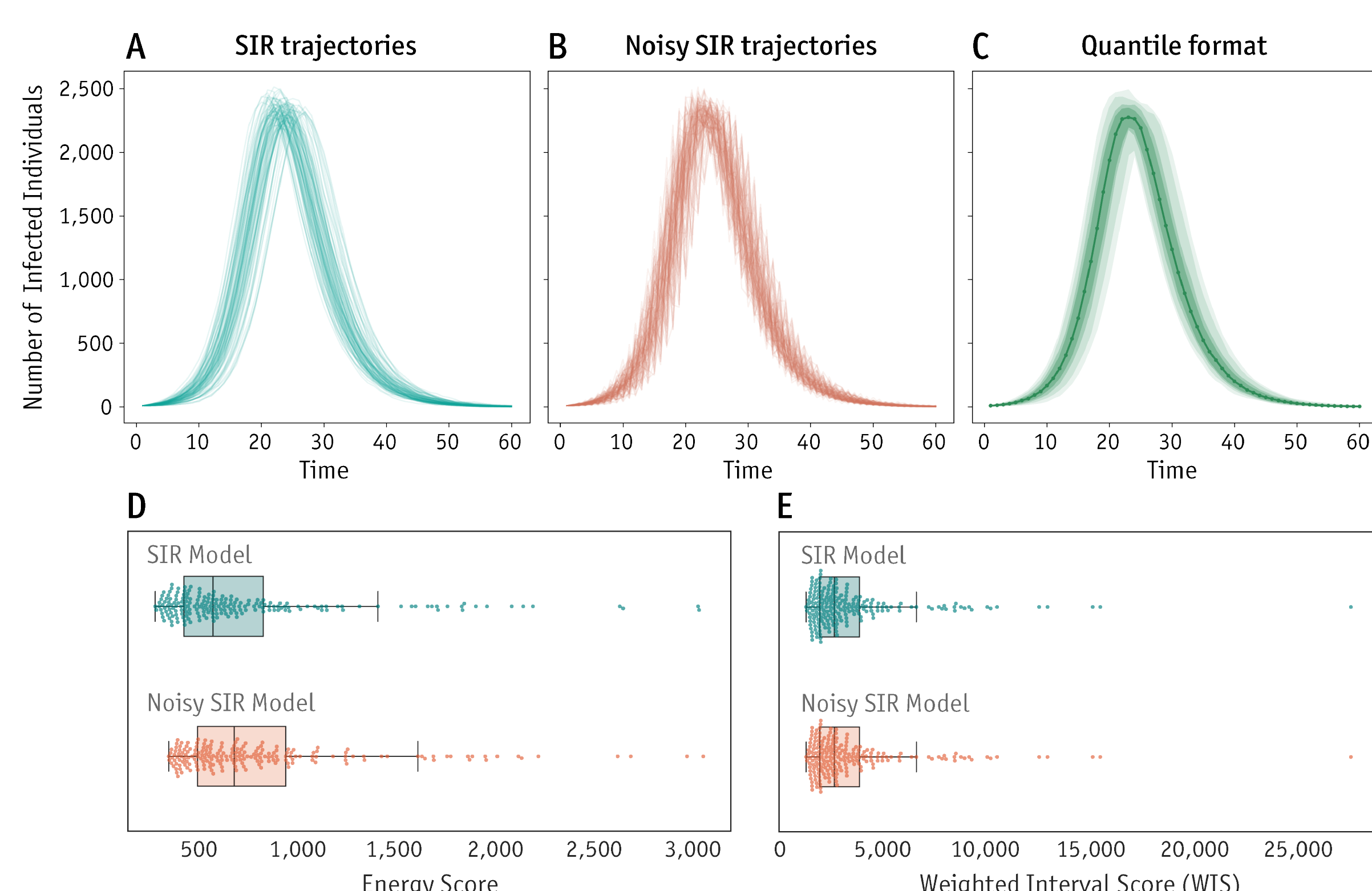
$$ES_{\text{dim}}(A, Y) = \frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{j=1}^T \Phi_j^2 \sum_{m=1}^M (a_{jm}^{(i)} - y_{jm})^2} - \frac{1}{2N^2} \sum_{i=1}^N \sum_{k=1}^N \sqrt{\sum_{j=1}^T \Phi_j^2 \sum_{m=1}^M (a_{jm}^{(i)} - a_{jm}^{(k)})^2}$$

$$ES_{\text{norm}}(P, y) = \frac{ES(P, y)}{\sum_{m=1}^M y_m}$$

## Advantages of the Energy Score

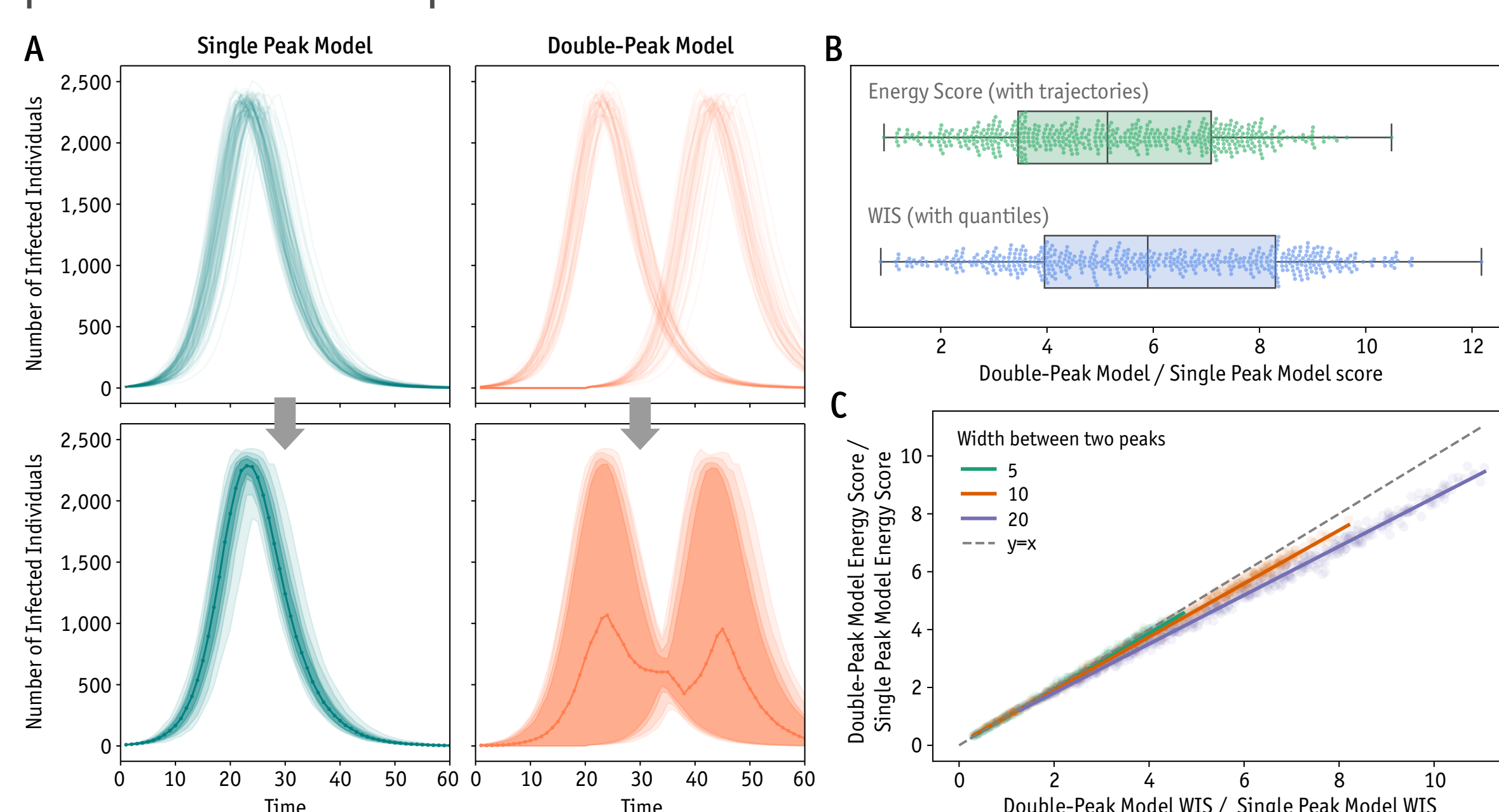
### Proper vs. Strictly Proper Scoring Rules.

- We generate trajectories from 2 stochastic epidemic models with the same marginal distribution at each time step and compare scores from a commonly used proper score applied to quantiles, the weighted interval score (WIS), against the strictly proper ES.
- The ES distinguishes between the 2 processes, but the WIS scores them identically.



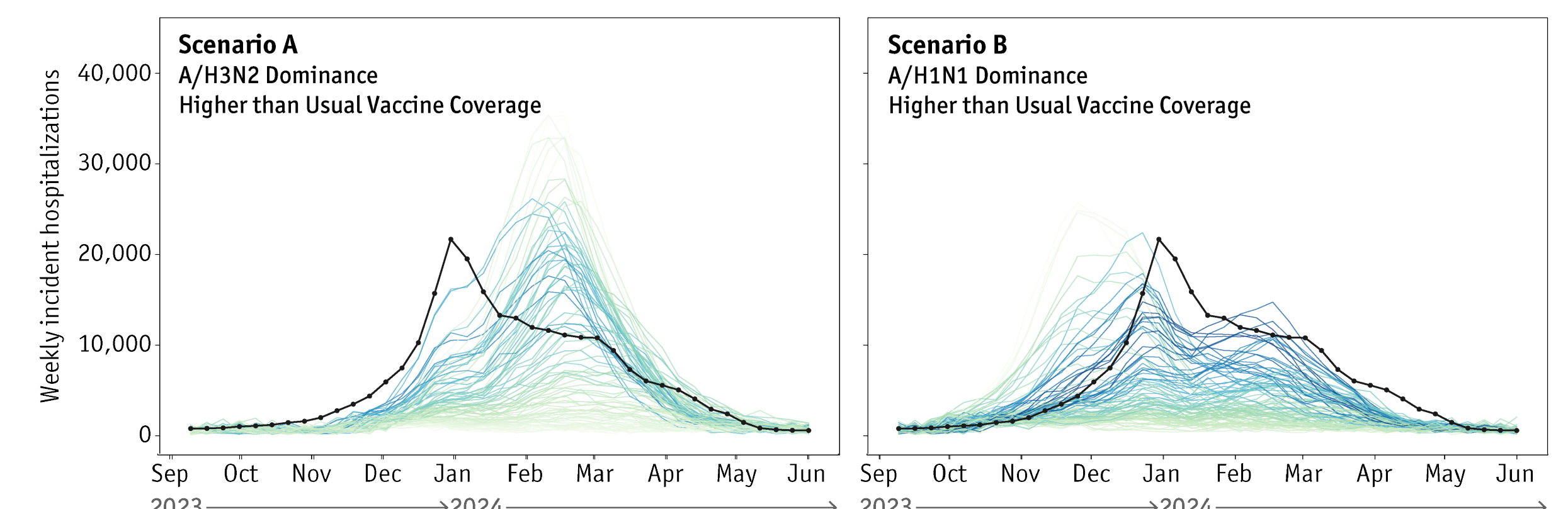
### Forecast Models with Multiple Peak Timing

- We produce 2 SIR models, one with a single peak and one with 2 peaks where each trajectory has a single peak but across the ensemble peaks are at 2 different times, comparing scores given by the ES and WIS.
- The WIS penalizes the double-peak model more than the ES. This discrepancy increases as the two peaks become farther apart.

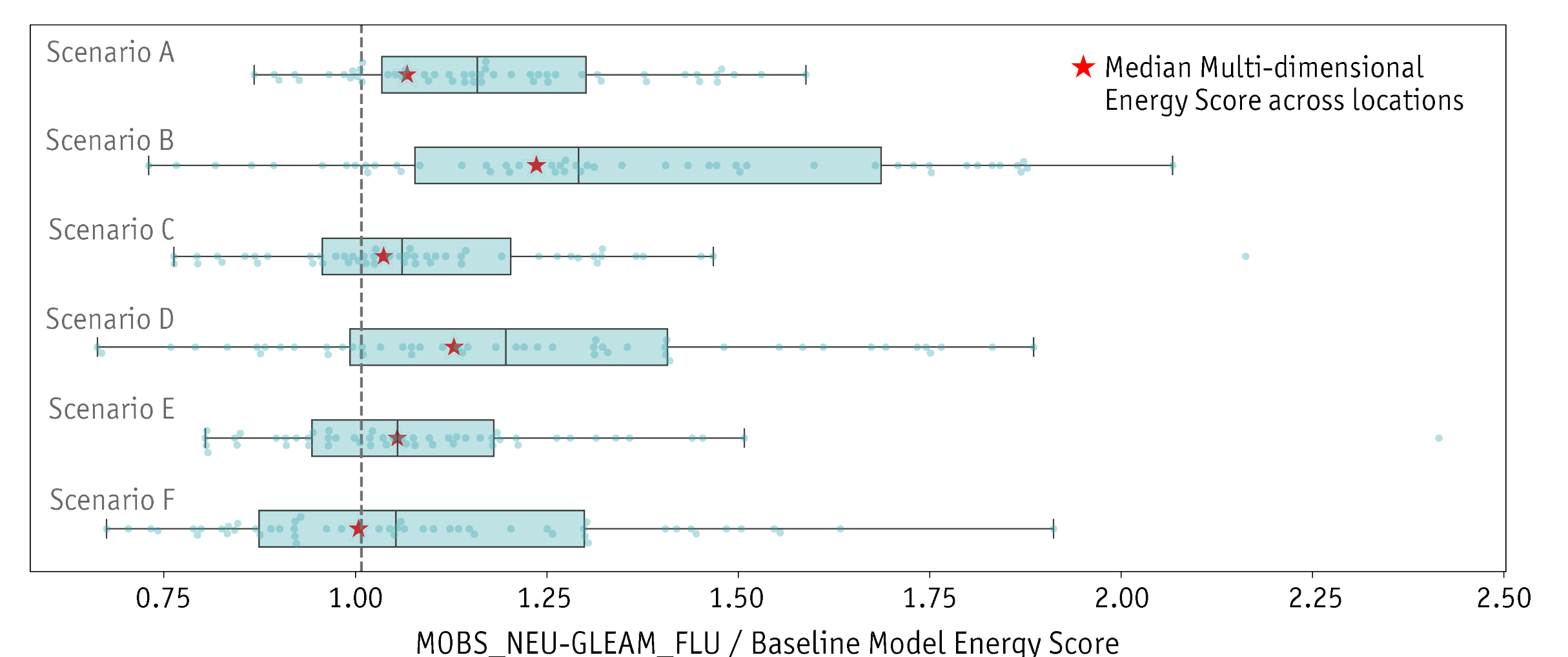
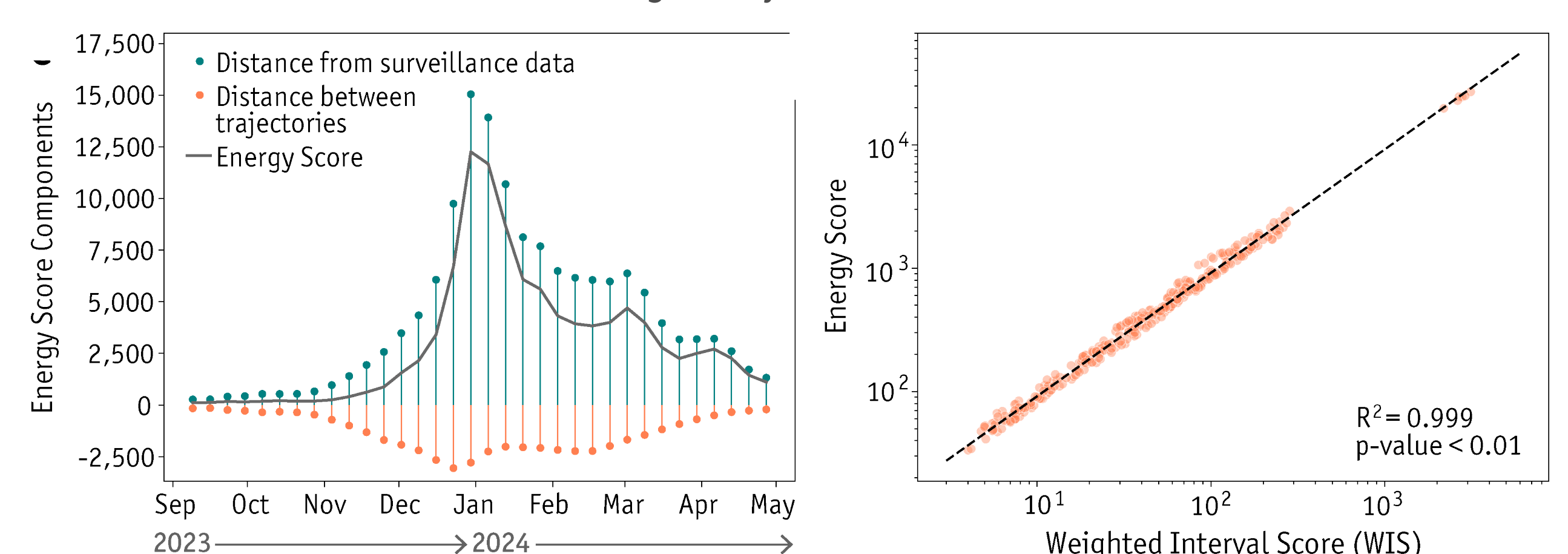


## Case Study from the Flu Scenario Modeling Hub

- Apply energy score to **2023-24 round of Flu Scenario Modeling Hub**, which explores the impact of vaccine coverage levels and the dominant circulating strain on weekly hospitalizations for U.S. states and nationally. We focus here only on the MOBS NEU-GLEAM FLU model.



- We show how the **ES depends on the magnitude** of the surveillance signal, especially near the peak, and we find that the **ES and WIS are highly correlated**.
- We rank the scenarios against each other and find similar ranking across the distribution of ES ratios across locations with the value given by the multi-dimensional ES.



## Discussion

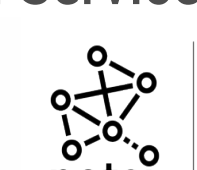
- The ES provides a framework for evaluating epidemic model outputs in **trajectory format**, avoiding the need of the additional aggregation step to summarize stochastic realizations into quantiles, preserving information that would be partially lost in fixed-time descriptive statistics.
- The ES distinguishes between predictions with identical marginal distributions but different underlying generative processes, thus more **accurately assessing complex features** such as multi-peak epidemic dynamics.
- We **introduce a multi-dimensional ES** that enables a comprehensive evaluation of model performance across multiple targets or locations simultaneously.
- One limitation of the ES is the increasing **computational cost** of the energy score with the number of trajectories. Randomly sampling a small percent of trajectories gives good agreement with the ES found through using the full set of trajectories.
- We consider equally weighted trajectories, but the ES allows **trajectory weights** to be individually defined, which opens the door to scoring that adjust weights based on past performance or specific outcomes.
- Beyond the case study of the Flu Scenario Modeling Hub, the ES is broadly **applicable to any epidemic forecast or prediction** that produces stochastic realizations.

## Acknowledgements

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